

INTERNATIONAL COORDINATION OF TRADE AND GREEN INDUSTRIAL POLICIES

Juan Dodyk

Washington University in St. Louis

INTRODUCTION

I study **two climate policies**:

- **Carbon pricing** or regulations: polluter pays. Example: EU ETS.

More efficient: it minimizes the social cost of achieving a given reduction in carbon emissions.

- **Green industrial policy** (GIP): subsidize investment in clean production.
Example: Inflation Reduction Act of 2022.

Politically useful: it grows the group of green producers, which helps to balance the opposition of dirty producers.

My JMP studies the *dynamic* linkage between them: you use GIP first to build a green coalition, and then you can use carbon pricing.

THIS PAPER

International spillovers. In a textbook new trade theory model,

- GIP at Home attracts clean entry, but *discourages clean entry abroad*. I call this the **green relocation effect**.
- Thus, relaxing political constraints at Home tightens political constraints abroad. GIP can have **negative political spillovers**.
- Also, because of the green relocation effect, there is a **protectionist motive for using climate policy**, green subsidies in particular. (In fact, they are illegal under the WTO.)

QUESTIONS

1. If countries cooperate both on trade and climate policies, what is the optimal agreement?

Answer: free trade, a large carbon tax, **and green subsidies** if political constraints bind (implication: the WTO should legalize but regulate them).

2. If countries choose both trade and climate policies non-cooperatively, what is the equilibrium?

Answer: they use carbon taxes, import tariffs and export subsidies, but they don't use green subsidies, unless political constraints bind. **GIP is second best both as a climate *and* as a trade policy.**

3. If countries cooperate on trade policy (ignoring climate), but choose climate policies non-cooperatively, what is the equilibrium climate policy?

Answer: there is a **green subsidy race**. Both countries use green subsidies, even if political constraints don't bind (in fact, even if they don't care about climate!).

THE MODEL – PRODUCTION

Two countries: H and F (Home and Foreign), endowed with labor L_i .

Manufacturing varieties $v \in [0, 1 + g_i]$ in each country. Producers are monopolists and consumers value variety.

- A fixed mass of incumbent varieties, $v \in [0, 1)$, uses a **dirty technology** with labor cost $l = q$.
- New **clean producers**, $v \in [1, 1 + g_i]$, enter after paying a fixed cost $f > 0$. Their labor cost is $l = f + cq$.
- Free entry implies zero profits for the marginal clean producer.

Varieties are traded with **iceberg transport cost**: to consume q units abroad we have to send θq units, with $\theta > 1$.

There is a **numéraire good** N , which is freely traded and produced with cost $l = q$.

THE MODEL – CONSUMPTION

Consumer payoff (and welfare) in country j is

$$\mathbf{W}_j = \log \left[\left(\sum_{i=H,F} \int_0^{1+g_i} q_{ij}(v)^{1/\mu} dv \right)^\mu \right] + N_j - D_j,$$

where $\mathbf{q}_{ij}(\mathbf{v})$ the quantity of variety v from i consumed in j , $\mu \equiv \frac{\sigma}{\sigma-1}$, $\sigma > 1$ is the elasticity of substitution of varieties, $\mathbf{N}_j$ is the quantity of the numéraire consumed in j .

Climate damages are

$$\mathbf{D}_j \equiv \gamma \sum_{i=H,F} \int_0^1 q_i(v) dv,$$

where $\mathbf{q}_i(\mathbf{v}) \equiv q_{ii}(v) + \theta q_{ij}(v)$ is total production of incumbent dirty variety v in country i , and γ measures the **social cost of carbon**.

THE MODEL – POLICIES

Country i imposes a **carbon tax** τ_i paid per unit of v produced with dirty technology.

Green investment subsidy ϕ_i paid to clean entrants.

Ad-valorem **tariffs** t_{ji} and **export subsidies** s_{ij} for both $j = H, F$.

Equilibrium consumer **prices** are

$$p_{ij}(v) = (1 + t_{ij})\theta^{1_{i \neq j}} \mu c_i(v) / (1 + s_{ij}),$$

where marginal cost $c_i(v)$ is c for clean varieties and $1 + \tau_i$ for dirty varieties.

A clean entrant's **profits** are

$$\pi_i(v) = (\mu - 1)cq_i(v) - (f - \phi_i).$$

Free entry determines g_i through $\pi_i(1 + g_i) = 0$.

Uniform **lump-sum transfer** to consumers to balance the budget.

THE MODEL – POLITICAL CONSTRAINTS

Status quo: no climate policy. Policymakers maximize welfare W_i , but a policy change requires support from at least n_i manufacturing producers.

Timing:

1. Policymakers propose (τ_i, ϕ_i) .
2. Producers vote. A proposal passes if at least n_i support it.
3. Clean producers invest and enter; production and consumption occur.

Producers care about profits.

- Dirty incumbents oppose climate policy because it reduces their profits.
- Clean entrants support it and earn zero profits by free entry.

Political constraint: the number of clean entrants must satisfy $g_i \geq n_i$.

FIRST QUESTION

Suppose that H and F choose both trade and climate policies in a trade agreement that is subject to political constraints. Ignore political constraints. What do they do?

Easy result: the optimal international trade-climate agreement involves

- uniform consumption subsidies $t_{ij} = -1/\sigma$ (to undo the monopoly markup),
- no (additional) import tariffs or export subsidies,
- carbon taxes equal to the global social cost of carbon, $\tau_i = 2\gamma$,
- green subsidies only if political constraints $g_i \geq n_i$ bind.

Key takeaway: GIP is the right policy instrument to overcome political constraints. This matters, because the WTO doesn't allow this kind of GIP (for good reasons, as we'll see).

SECOND QUESTION

Suppose that H and F choose trade and climate policies non-cooperatively. Ignore political constraints. What do they do?

I can show that in an interior equilibrium there are numbers $\{\eta_i, \lambda_i\}_{i=H,F}$ such that for all $i, j \in \{H, F\}, i \neq j$:

$$t_{ij} = \begin{cases} \frac{\gamma}{\mu(1+\gamma)}(1 + s_{ij}) & \text{for dirty varieties,} \\ \frac{\eta_i}{\sigma g_i}(1 + s_{ij}) & \text{for clean varieties,} \end{cases}$$

$$t_{ii} = -1/\sigma,$$

$$s_{ij} = \lambda_j P_j^{1-\sigma}(1 + t_{ij}),$$

$$\tau_i = \gamma,$$

$$\phi_i = 0.$$

Trade policy is used for several reasons: carbon leakage, TOT manipulation, and exploiting the green relocation effect. But **green subsidies are not needed!**

THIRD QUESTION

Suppose that H and F sign a trade agreement ignoring climate, so $t_{ij} = -1/\sigma$ and $s_{ij} = 0$. Then they choose climate policies τ_i, ϕ_i non-cooperatively.

PROPOSITION

If political constraints don't bind and θ is large, in equilibrium $\tau_i < \gamma$ and $\phi_i > 0$.

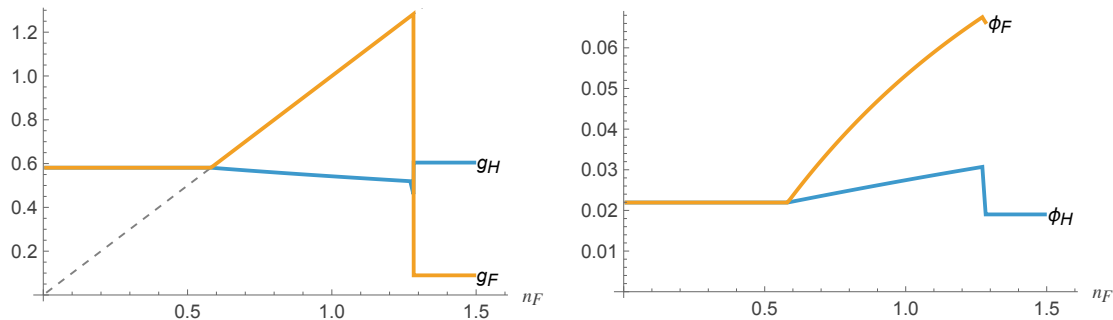
Why do countries use green subsidies? To exploit the **green relocation effect**: green subsidies at Home move green production (which is more efficient) to Home, saving on transport costs.

Foreign has the same incentive, and a **subsidy race** takes place.

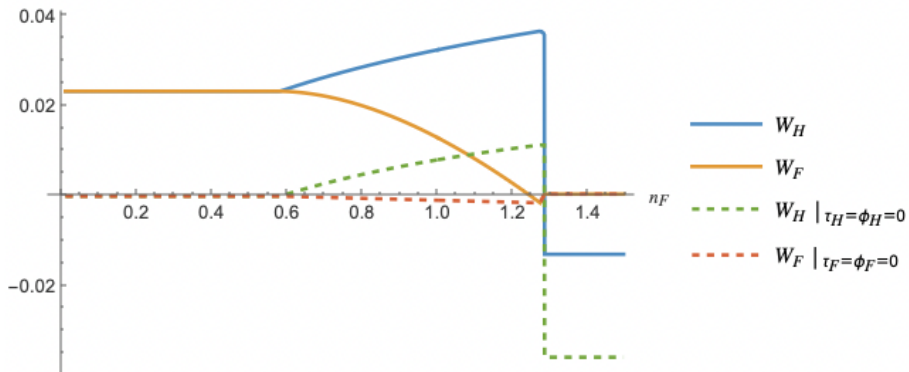
Carbon taxes are lower than the SCC as a result, because green subsidies are doing some emissions reductions.

WHAT HAPPENS IF POLITICAL CONSTRAINTS BIND?

Suppose that $n_H = 0$ and we increase n_F . At first $g_F \geq n_F$ doesn't bind, but at some point it does.



When $g_F \geq n_F$ binds, both countries spend more on green subsidies, but g_H decreases. Eventually it's no longer worth it for Foreign, and $\tau_F = \phi_F = 0$. Home reduces ϕ_H , but g_H jumps.



Political constraints at F reduce welfare for F . When they bind, they increase W_H . But when F “gives up”, overall welfare goes down.

MORE STUFF

If both political constraints bind countries may be forced to play mixed strategies.

If one country can play first and commit to climate policy, maybe they can leverage the other country's political constraints and **deter** them from implementing any policies (and, thus, from entering the subsidy race).

CONCLUSION

- **Conceptual point:** green industrial policy may have positive political effects at home, but *negative political externalities* abroad.
- The literature has studied the “reverse leakage” effect of green subsidies (Kotchen & Maggi 2025), by which domestic subsidies reduce emissions abroad. In my model subsidies also have a **green relocation effect**, by which they *increase* emissions abroad.
- **Subsidy races** can occur. Countries may use subsidies for **deterrence**. China?
- **Normative takeaway:** the optimal trade-climate agreement **doesn't ban discriminatory green subsidies**, but does regulate them.
 - Existing environmental provisions only coordinate trade policy and environmental regulation.
 - International coordination of green industrial policy can be beneficial.
- Work in progress. Many natural questions I still haven't answered.